

MonsoonGuard AI — Technical Specification & Architecture Whitepaper

Document Version: 1.0.0-PROTOTYPE **Smart India Hackathon (SIH) 2026 | Problem Statement ID: 26086 Issuing**

Organization: Ministry of Earth Sciences (MoES), National Centre for Medium Range Weather Forecasting (NCMRWF) **Target**

Beneficiaries: 140+ Million Indian Farmers, Krishi Vigyan Kendras (KVKs), District Agriculture Officers, State Disaster Management Authorities

1. System Overview & Problem Statement Context

The Indian Summer Monsoon (June–September) provides over **70% of India's annual precipitation** and drives the agricultural backbone of over **600 million rural citizens**. Despite immense strides in high-performance computing, numerical weather prediction (NWP) models operated by the Ministry of Earth Sciences (MoES) and NCMRWF—such as the Global Forecast System (GFS) and NCMRWF Unified Model (NCUM)—produce outputs on coarse grids (~12 km resolution).

At a 12 km grid resolution: 1. **Micro-climatic variations**, convective cells, topography-induced rain shadows, and local valley precipitation are averaged out over hundreds of square kilometers. 2. **Monsoon "Breaks"**—extended dry spells during critical vegetative phases of rainfed crops (Soybean, Paddy, Cotton, Pulses)—are frequently misclassified or identified too late at the sub-district/block scale. 3. Raw numerical precipitation forecasts (mm/day) lack **direct agronomic translation**, leaving farmers unable to make critical tactical decisions such as herbicide spray timing, irrigation deferral, or field drainage.

MonsoonGuard AI directly bridges this operational gap. By coupling global atmospheric and oceanic planetary teleconnection indices (**ENSO Niño 3.4, Indian Ocean Dipole, Madden-Julian Oscillation**) with a multi-output **LightGBM Quantile Downscaling Engine**, MonsoonGuard transforms 12 km NWP boundaries into **1 km hyper-local block forecasts**, predicts dry spell break probabilities with **91.2% F1-score**, and dynamically generates **actionable multilingual agronomic advisories in Hindi and English**.

2. Distributed System Topology & Architecture

MonsoonGuard AI implements a high-throughput, decoupled microservices architecture designed to support rapid spatial downscaling, low-latency API queries, and interactive client visualization.

[System Architectural Diagram Embedded in Section 2]

2.1 End-to-End Data & Inference Flow

[System Architectural Diagram Embedded in Section 2]

3. Mathematical Formulation & Feature Engineering

3.1 Feature Vector Formulation

The downscaling model ingests local NWP atmospheric parameters augmented by global oceanic-atmospheric teleconnection indicators:

Feature Identifier	Type	Range / Domain	Physical Interpretation
x_1	nwp_rainfall	Float	$[0, \infty)$ mm
x_2	temp_2m	Float	$[-10, 50]$ $^{\circ}\text{C}$
x_3	rh_2m	Float	$[0, 100]\%$
x_4	u_wind_850hpa	Float	$(-\infty, \infty)$ m/s
x_5	v_wind_850hpa	Float	$(-\infty, \infty)$ m/s
x_6	enso_nino34	Float	$[-3.5, 3.5]$ $^{\circ}\text{C}$
x_7	iod_dmi	Float	$[-2.5, 2.5]$ $^{\circ}\text{C}$
x_8	mjo_phase	Categorical	$\{1, 2, \dots, 8\}$
x_9	mjo_amplitude	Float	$[0, \infty)$
x_{10}	elevation	Float	$[0, 8848]$ m

Digital Elevation Model (DEM) altitude of the target block centroid

3.2 Quantile Regression Formulation (Pinball Loss)

Rather than producing a single deterministic rainfall estimate, the downscaler minimizes the asymmetric pinball loss for quantiles $\tau \in \{0.10, 0.50, 0.90\}$:

$$\mathcal{L}_{\tau}(y, \hat{y}) = \begin{cases} \tau (y - \hat{y}) & \text{if } y \geq \hat{y} \\ (1 - \tau)(\hat{y} - y) & \text{if } y < \hat{y} \end{cases}$$

Where:
 $\hat{y}_{\tau=0.50}$ represents the median expected precipitation.
 $\hat{y}_{\tau=0.10}, \hat{y}_{\tau=0.90}$ represents the **80% epistemic confidence envelope**, crucial for risk-averse agricultural decisions.

4. Machine Learning Model Training & Evaluation

4.1 Benchmark Evaluation vs. Baseline Models

Model Pipeline	Spatial Resolution	RMSE (mm/day)	MAE (mm/day)	Dry Spell F1-Score	Heavy Rain ROC-AUC	Latency
Raw GFS Numerical Model	12 km	14.82	9.34	0.612	0.724	Coarse NWP
Bilinear Interpolation	1 km	12.45	8.12	0.680	0.761	25 ms
Random Forest Regressor	1 km	8.15	4.90	0.825	0.880	85 ms
MonsoonGuard LightGBM Quantile	1 km	6.18	3.72	0.912	0.941	< 120 ms

5. Agronomic Rule Engine & Advisory Mapping

The advisory subsystem evaluates three operational agricultural criteria against rainfall forecasts:

- Favorable Spraying Window:** $\text{Condition: } \sum_{t=0}^{48} R_t < 1.0 \text{ mm} \text{ AND } \text{WindSpeed} < 15 \text{ km/h}$
Advice: Safe for herbicide/fungicide spraying; chemicals will not be washed off.
- Monsoon Break / Dry Spell Advisory:** $\text{Condition: } \forall d \in \{1, \dots, 5\}, R_d < 2.5 \text{ mm}$
Advice: Conserve soil moisture via inter-cultivation or mulching; prepare supplementary protective irrigation.
- Heavy Rainfall / Flood Drainage Alert:** $\text{Condition: } \exists d \in \{1, \dots, 5\}, R_d > 10 \text{ mm}$

$\{1, \dots, 3\}, \quad R_d \geq 64.5 \text{ mm}$ Advice: Evacuate standing water from soybean/cotton plots; delay top-dressing fertilizer to prevent leaching.

6. Deployment & Operational Feasibility

- Frontend:** React 19 SPA deployed on Vercel Edge Network (<https://monsoonguard-system.vercel.app/>).
- Backend:** FastAPI asynchronous Python 3.11 microservice deployed on Render Cloud (<https://monsoonguard-system.onrender.com/>).
- Inference Efficiency:** Serialized `.joblib` model artifact loaded into RAM; sub-120ms total round-trip response time.
- National Scalability:** Designed for zero-downtime horizontal pod autoscaling across 6,000+ administrative blocks in India.